Bridging Product Catalogs and Customer Reviews for E-Commerce via Multi-Aspect Joint Retrieval*

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Abstract

E-commerce platforms comprise multiple retrieval needs, from locating relevant products to surfacing useful information within customer reviews. Relevance in these tasks depends not only on textual content but also on structured aspects like brand, category, price, rating, and popularity. While multi-aspect dense retrievers have shown that such signals boost product search performance, existing models are typically built for a single task and corpus, leaving unclear whether aspect-aware representations can generalize across related e-commerce retrieval tasks. We address this gap by studying joint multi-aspect dense retrieval, where one model handles both product and review search. We introduce J-MADRAL, a joint retrieval model that learns shared aspect-aware representations from text and metadata across both tasks. To support evaluation, we construct Search ESCI, a new synthetic benchmark for review search built from Amazon Reviews 2023 and ESCI product resources, and test our approach on Amazon ESCI, TREC Product Search 2023, and Search ESCI. Compared to sparse, dense, and multi-aspect baselines, joint training yields substantial gains in review search while remaining competitive on product search, demonstrating the benefit of shared aspect-aware representations for unified e-commerce retrieval. Source code, data, and model weights are available at https://github.com/alemarco96/J-MADRAL.

Keywords

Product Search, Review Search, Joint Multi-Aspect Dense Retrieval

1. Introduction

E-commerce search extends beyond retrieving products from a catalog. Satisfying users' requests often demands evidence from multiple sources, including product descriptions and structured metadata, but also customer reviews discussing product quality, usage scenarios, defects, or particular product characteristics. Therefore, to support such an integrated search experience, retrieval models are required to operate across heterogeneous yet closely related e-commerce corpora. Product search is closely related to Information Retrieval (IR), while being distinguished from classical ad-hoc search and passage retrieval due to the nature of the searchable units. The catalog items comprise textual content and are enriched with structured metadata organized into aspects (brand, category, price, color, popularity) drawn from finite vocabularies. Such data could be leveraged to improve product representations and boost retrieval effectiveness by framing product search as a multi-aspect retrieval task. Before Pretrained Language Models (PLMs), successful approaches included latent Dirichlet allocation [2] and vector space models [3, 4]. More recently, neural methods have used aspect extraction and fusion to build aspect-aware product representations, either through dedicated fusion networks [5] or by treating aspects as auxiliary text and optimizing bidirectional aspects-content objectives [6]. However, these approaches target a single task, product retrieval, and leave open whether aspect-aware representations can transfer across related e-commerce tasks, including product and review search, despite both reflecting user engagement with products. This perspective connects with recent work on unifying search and recommendation, which has shown that shared representations and cross-task

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signals benefit both product retrieval and preference modeling tasks [7, 8, 9, 10, 11, 12]. Instead, our setting focuses on jointly training retrieval models across heterogeneous e-commerce corpora. To address this gap, we propose J-MADRAL, an aspect-aware dense retrieval framework jointly trained on different corpora, products and customer reviews. They share domain vocabulary, aspect semantics, and underlying user intent, enabling J-MADRAL to learn a shared representation space combining catalog-level aspect signals with review-level evidence to support both tasks. Our contributions are threefold: (i) we formulate joint aspect-aware retrieval over product catalogs and customer reviews as an integrated e-commerce search problem; (ii) we propose J-MADRAL, extending a MADRAL-inspired multi-aspect dense retriever to heterogeneous e-commerce corpora; and (iii) we construct a novel review-search benchmark from Amazon reviews and empirically evaluate the proposed setting. We address the following research questions: *RQ1*: Can multi-aspect dense retrieval extend effectively from product to review search? *RQ2*: Does joint training over both product and review corpora outperform task-specific training? *RQ3*: Does combining multi-aspect learning with joint training improve the balance between review-search gains and product-search effectiveness compared to either strategy applied in isolation? The results show that joint aspect-aware training substantially improves review retrieval while remaining competitive on product search, suggesting to view these tasks not as independent tasks but as complementary parts of a unified e-commerce search system.

2. Methodology

In multi-aspect dense retrieval, queries and documents combine textual content with structured metadata organized into aspects $\mathscr{A} = a_1, \ldots, a_{|\mathscr{A}|}$, each drawn from a finite vocabulary $\mathscr{V}_{a_k}$. By incorporating this metadata into query and document representations, such models capture fine-grained relevance signals beyond textual semantic similarity. Building on MADRAL [5], J-MADRAL introduces three adaptations for joint e-commerce retrieval: (i) a shared representation space for products and reviews; (ii) the unsupervised *other* aspect is replaced with the encoder's `[CLS]` embedding, removing the need for extra aspect supervision; and (iii) an aspect-gating fusion mechanism that learns the relative contribution of each aspect to the final representation. Following prior work [13, 6, 14], the model employs BERT [15] as its backbone encoder and incorporates an Aspect Extraction Network (AEN) module to derive aspect-specific embeddings through self-attention. Attention queries are input-independent trainable parameters (one per aspect), allowing the model to selectively attend to the most relevant parts of the input text for each aspect. Salient information not covered by explicit aspects is captured by the encoder's `[CLS]` embedding. This design choice is well aligned with the pre-training objective of BERT and other PLMs, as it yields a global sequence representation without requiring additional aspect supervision or large-scale fine-tuning. The Aspect Fusion Network (AFN) module fuses multiple aspect embeddings into a single representation for efficient retrieval. As none of the fusion strategies proposed in the original paper performed consistently well in preliminary experiments, we introduce an *aspect-gating* weighting scheme in which aspect embeddings both encode semantic information and estimate their own relative importance. A shared linear projection $\mathbf{W}_g \in \mathbb{R}^{H\times 1}$ computes an importance score for each aspect $\mathbf{w} = \sigma(\mathbf{A}\mathbf{W}_g)$, where σ is softmax over the aspects $\mathscr{A}$, and the final representation is the resulting weighted average of aspect embeddings.

Training Procedure J-MADRAL is trained in two stages: *pre-training* and *fine-tuning*. During pre-training, the model is adapted to the target domain using the standard Masked Language Model (MLM) objective complemented by an auxiliary *Aspect Prediction (AP)* loss. For each aspect $a_k \in \mathscr{A}$, the corresponding embedding $\mathbf{a}_k$ is projected through an aspect-specific prediction layer $\mathbf{W}_k \in \mathbb{R}^{H\times|\mathscr{V}a_k|}$ to predict its ground-truth value $v^* \in \mathscr{V}a_k$. The AP loss is defined as:

$$\mathscr{L}_{AP} = \sum_{\mathbf{a}_k \in \mathscr{A}} -log\left(\frac{exp(\mathbf{a}_k \cdot \mathbf{W}_k[v^*])}{\sum_{v\in\mathscr{V}a_k} exp(\mathbf{a}_k \cdot \mathbf{W}_k[v])}\right) \tag{1}$$

The pre-training objective combines both losses using a linear scaling factor: $\mathscr{L}_{pre-training} = \mathscr{L}_{MLM} + \lambda_p \mathscr{L}_{AP}$, mixing queries, items, and reviews from both datasets. During fine-tuning, we construct task-specific mini-batches and alternate between product and review search batches, using gradient accumulation over two steps. This allows each parameter update to incorporate gradients from both tasks while preserving task-specific negatives, as in-batch negatives are only valid within a single retrieval task. The fine-tuning linearly combines the relevance cross-entropy loss with aspect prediction loss: $\mathscr{L}_{fine-tuning} = \mathscr{L}_{CE} + \lambda_f \mathscr{L}_{AP}$.

3. Experimental Settings and Results

Given the absence of public datasets supporting both retrieval tasks within the same domain, we construct a unified collection from multiple sources. *Amazon ESCI* [16] [1] is a well-known large-scale dataset for product search. Its English portion contains 1.2M distinct products, enriched with metadata from Amazon Reviews 2023 [17] and ESCI-S, including *brand*, *categories* (first four hierarchy levels), *color*, *popularity* (number of ratings), and *price*. The training, validation, and test sets contain 74.9K, 13.5K, and 8.9K queries, respectively. Relevance assessments use four labels, *Exact*, *Substitute*, *Complementary*, and *Irrelevant*, with assigned gains of 1.0, 0.1, 0.01, and 0.0. For fine-tuning and binary metrics, only *Exact* products are considered positive. *Search ESCI* is a novel synthetic dataset for review search and one of the contributions of this work. Leveraging Amazon Reviews 2023 [17] and ESCI-S, we extract a corpus of 22.1M reviews of at least 200 characters annotated with product-inherited *categories*, 1-5 star *ratings*, and *helpful votes* (number of users marking the review as helpful). Inspired by DataMorgana [18], we employ Qwen 3.5 9B to generate search-style queries from customer reviews, followed by rule-based filtering to remove malformed, overly vague, and non-self-contained queries. The dataset contains 75K, 12.5K, and 12.5K queries for training, validation, and testing, respectively. As in MS-MARCO [19], each query is paired with a single positive review, the one used to generate it. *TREC Product Search 2023* [20] is a product-search test collection based on Amazon ESCI corpus, comprising 186 test queries with extensive relevance assessments (620.66 judgements per query) following a four-level scale: *Perfectly Relevant*, *Highly Relevant*, *Related*, and *Irrelevant*, with gains of 3, 2, 1, and 0, respectively. For binary metrics, labels ≥ 2 are treated as positive, while 1 and 0 as negative.

Baselines We compare J-MADRAL against five baselines: (i) BM25 [21], a strong sparse bag-of-words retrieval model based on lexical matching; (ii) DRAGON [22], a state-of-the-art dense retriever for passage retrieval over unstructured text not specifically tailored to e-commerce; (iii) BiBERT [23], a dense retrieval model pre-trained with an MLM objective, using the `[CLS]` embedding as the final representation; (iv) ATTEMPT [6], which prefixes aspect labels to query/product text and is pre-trained with auxiliary bidirectional "aspects-content" mutual prediction MLM losses; (v) MURAL [14], which represents each aspect with multiple tokens at different granularities (phrase, word, and token level). For BiBERT and MADRAL, we evaluate three training settings: product-only (P), trained on Amazon ESCI; review-only (R), trained on Search ESCI; and joint (J), trained on both datasets.

Implementation Details We implemented all neural models using PyTorch [24] and Transformers [25], while BM25 results were obtained with Pyserini [26] ($k_1 = 1.2$, $b = 0.75$). Since queries lack aspect annotations, we automatically derive labels for training queries: for each aspect, a query receives the most frequent label among its positives only if it accounts for more than 50% of them; otherwise, no label is assigned. For Search ESCI, *rating* and *helpful votes* are excluded because they cannot be reliably inferred from query text. Product inputs concatenate titles, descriptions, and bullet points, while review inputs combine review titles and text; all inputs are truncated to 128 tokens. The models are shared across queries and items/reviews to improve knowledge sharing, initialized from BERT, and trained for 20 epochs with Adam [27] and linear warm-up. Pre-training uses a learning rate of 1×10^{-4}, an

[1] Amazon ESCI: https://github.com/amazon-science/esci-data; ESCI-S: https://github.com/shuttie/esci-s

Table 1
Product and review search results. We highlight the best and second-best results in bold and underline, respectively. † denotes significant differences with respect to J-MADRAL (paired t-test, $p < 0.05$, using Bonferroni correction). Results for ATTEMPT and MURAL on Amazon ESCI are taken from prior work [6, 14].

	Search ESCI				Amazon ESCI				TREC Product Search 2023			
	R@100	R@500	MRR	nDCG@50	R@100	R@500	MRR	nDCG@50	R@100	R@500	MRR	nDCG@50
BM25	0.5875†	0.7288†	0.2539†	0.3101†	0.4949†	0.6603†	0.4056†	0.3160†	0.7252	0.8650	0.7746†	0.5988†
DRAGON	0.5451†	0.6751†	0.2347†	0.2873†	0.5490†	0.7155†	0.4541†	0.3540†	0.7373	0.8692†	0.8135	0.6293
ATTEMPT [6]	—	—	—	—	0.6233	0.7924	—	0.4097	—	—	—	—
MURAL [14]	—	—	—	—	**0.6371**	**0.8023**	—	**0.4228**	—	—	—	—
R-BiBERT	0.5972†	0.7350†	0.2602†	0.3176†	—	—	—	—	—	—	—	—
R-MADRAL	0.6405†	0.7626	0.2944†	0.3541†	—	—	—	—	—	—	—	—
P-BiBERT	—	—	—	—	0.6018†	0.7640†	0.4939	0.3967†	0.7432	0.8757	0.8256	0.6287
P-MADRAL	—	—	—	—	0.6235†	0.7806†	**0.5060**	0.4104†	**0.7547**	0.8840	**0.8337**	**0.6480**
J-BiBERT	0.6300†	0.7593†	0.2879†	0.3474†	0.5889†	0.7556†	0.4808†	0.3858†	0.7432	0.8684†	0.8114	0.6213
J-MADRAL	**0.6488**	**0.7729**	**0.3007**	**0.3611**	0.6083	0.7729	0.4988	0.4028	0.7527	**0.8888**	0.8333	0.6411

MLM masking ratio of 0.15, and $\lambda_p = 0.1$. Fine-tuning uses a 5×10^{-6} learning rate, batch size 64, 7 hard negatives per query in addition to in-batch negatives, and $\lambda_f = 0.05$.

Results The overall results are reported in Table 1. J-MADRAL achieves strong performance across all datasets, matching competitive single-task models on product search while substantially improving review search, a more challenging task requiring bridging user needs, product metadata, and review evidence. On *Search ESCI*, aspect-aware representations provide substantial benefits: R-MADRAL improves over R-BiBERT by 7% in Recall (R) and 13% in Mean Reciprocal Rank (MRR) and Normalized Discounted Cumulative Gain (nDCG). Joint training yields further gains, with J-MADRAL achieving the best results overall; a similar pattern holds for J-BiBERT outperforming R-BiBERT despite lacking explicit aspect-aware representations. In contrast, BM25 and DRAGON perform less effectively, highlighting the importance of domain-specific training for e-commerce retrieval. These results indicate that product and review search provide complementary supervision, enabling knowledge transfer via joint training. On *Amazon ESCI*, P-MADRAL achieves the strongest results among our trained models, matching ATTEMPT [6] and approaching MURAL [14]. On *TREC Product Search 2023*, P-MADRAL and J-MADRAL perform similarly, with the former leading on ranking metrics and the latter on R@500. Joint training does not consistently improve product search effectiveness: J-MADRAL is slightly behind P-MADRAL on Amazon ESCI, while the two are closely matched on TREC Product Search 2023. This is likely because product search already benefits directly from product-level aspect supervision, while the review search objective introduces only partially aligned information as review-specific aspects like ratings and helpful votes may add noise to product ranking. Nevertheless, the limited impact on product search combined with strong gains on review search supports our central motivation: product and review retrieval are related e-commerce tasks that benefit from shared representation learning.

4. Conclusions and Future Work

In this paper, we introduced J-MADRAL, a unified aspect-aware model for joint product and review retrieval leveraging data across two related e-commerce tasks, and Search ESCI, a synthetic review search collection for evaluating retrieval of customer reviews containing evidence about product quality, properties, and usage. Our results show that the two tasks provide complementary supervision: joint training yields substantial gains for review search while preserving competitive product-search effectiveness. A limitation is that Search ESCI relies on synthetic queries and single positive relevance labels, potentially underrepresenting the diversity of relevant review evidence. These findings support integrated e-commerce retrieval models that jointly exploit catalog-level structure and review-level evidence to serve the diverse information needs of online shoppers. Future work will explore integrating joint architectures such as J-MADRAL into conversational systems supporting product search and related retrieval within the same framework, enabling richer user interactions.

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Declaration on Generative AI

Generative AI tools were used only to assist with language polishing and grammar checking. All research ideas, experimental design, implementation, analysis, and scientific claims were developed and critically reviewed by the authors.

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